SHAIQ BHATTI

AI/ML Engineer | Generative AI & Computer Vision
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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building and deploying machine learning, deep learning, Generative AI, RAG, and computer vision applications in Python. Experienced in developing end-to-end AI systems using PyTorch, TensorFlow, OpenCV, LangChain, LLMs, and FastAPI, from data processing and model training to evaluation and production deployment. Built real-time computer vision and document intelligence systems with measurable results, including 94% recall, 93.1% F1-score, and 35+ FPS processing.

CORE SKILLS

AI/ML : Machine Learning, Deep Learning, Generative AI, LLMs, RAG, LangChain, AI Agents, Prompt Engineering
Computer Vision : Object Detection, Object Tracking, Image Classification, OCR, YOLO, Image/Video Processing
Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, scikit-learn, OpenCV, NumPy, Pandas
Programming & Tools: Python, SQL, Git, C++, Data Structures & Algorithms
Backend & APIs: FastAPI, Django, REST APIs, PostgreSQL, MongoDB
NLP: LSTM, BiLSTM, NLP, TextBlob
Professional: Cross-Functional Collaboration, Researching & Evaluating New Techniques, Adaptability, Fast Learning, Teamwork

AI / ML PROJECTS

LEO — AI-Powered Home Assistant for the Elderly (Final Year Project)

- Led the computer vision design on a 3-person team to build LEO, a real-time, webcam-based fall-detection and care-assistant system for elderly users, owning the detection pipeline, backend, and conversational AI end-to-end.
- Designed, trained, and fine-tuned a dual-YOLO object detection and tracking pipeline in PyTorch with CUDA acceleration and OpenCV, processing live video at 35+ FPS on consumer-grade GPU hardware.
- Evaluated model performance against a 250-video real-world test set using precision, recall, and F1-score, then engineered a six-signal weighted verification algorithm (balance shift, body-outline change, motion blur, posture orientation, trajectory, and pose matching) with caregiver-configurable safe zones to suppress false positives — achieving 94% recall, 98% specificity, and a 93.1% F1-score.
- Optimized and deployed the trained models into a production FastAPI backend with PostgreSQL persistence, dispatching caregiver alerts in under 2 seconds of a confirmed fall.
- Developed "Leo," a Qwen-LLM-powered voice assistant with an intent classifier for medication reminders, emergency escalation, and everyday conversation with elderly users.
- Validated the system with elderly participants in daily real-world testing, achieving a 4.4/5 average user-satisfaction rating.

Multimodal Intelligence — Document Q&A and Reasoning Pipeline (Personal Project)

- Built an end-to-end retrieval-augmented generation (RAG) pipeline in Python that ingests PDF and text documents, extracts per-page text and tables, and threads page-level source locations through chunking, embedding, and retrieval.
- Implemented a local sentence-transformer embedding layer with a persisted Chroma vector store, plus a structured table registry for exact numeric lookups alongside semantic search.
- Designed a multi-stage reasoning engine that calls an LLM with a sandboxed, AST-whitelisted calculation tool for numeric reasoning, resolves inline citations back to exact source pages, and attaches a retrieval-based confidence score to every answer.
  Exposed the pipeline through a FastAPI backend (upload, query, and health endpoints) with a lightweight web frontend, and validated ingestion, extraction, an d retrieval ranking with an automated pytest suite.

AI-Powered OCR System

Built an OCR system in Python and OpenCV to automatically detect and extract text from images and scanned documents across various formats.

Object Detection System

Built a real-time object detection system using YOLO and pre-trained deep learning models to identify, classify, and track objects in video streams.

PROFESSIONAL EXPERIENCE

Junior AI Engineer Intern — BCG X **Jul 2026– Sep 2026**
Remote · Lahore, Pakistan

- Developed a Python-based financial analysis chatbot prototype that processes predefined natural-language queries and delivers financial insights for Apple, Microsoft, and Tesla, including revenue and net-income comparisons.
- Implemented query normalization, rule-based query handling, dynamic financial calculations, and response validation, with a six-case test suite covering supported and unsupported queries and all tests passing.

Machine Learning & AI Intern — The FastHire **Sep 2025 – Nov 2025**
Remote · Lahore, Pakistan

- Developed, trained, and evaluated machine learning models using core algorithms, optimizing predictive performance and accuracy across multiple datasets.
- Engineered computer vision applications with Python and OpenCV, implementing real-time image processing and object detection features.
- Built and tested full-cycle AI applications end-to-end, managing the workflow from data preprocessing and model training through optimization, deployment, and final project integration.

Data Analytics Intern — Deloitte **Jun 2025 – Aug 2025**
Remote · Lahore, Pakistan

- Analyzed large datasets using statistical methods to identify key trends and deliver actionable business insights.
- Built interactive Tableau dashboards to visualize data clearly and make findings accessible to cross-functional teams.
- Owned the end-to end data process, including cleaning raw data, performing analysis, and presenting final reports to stakeholders.

EDUCATION

BS Computer Science **2022 – 2026 (Completed)**
Government College University, Lahore, Pakistan

Intermediate in Pre-Engineering
Government College University, Lahore, Pakistan **2020 – 2022 (Completed)**

LANGUAGES

Urdu: Native | English: B2 Listening/Reading, B1 Speaking/Writing